



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University

Alangilan Campus
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College of Informatics and Computing Sciences

PROJECT PROPOSAL
OOP 101 - OBJECT-ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026-2027

I. TEAM FORMATION

Team Name:	Cup of Code		
Team Number:	6	Section:	2101

TEAM MEMBERS			
No.	SR-Code	Name	Github Username
1	25 - 02507	Arias, Jayro J.	jayro26
2	25 - 07443	Carenan, Ahron Simeon D.	Violet_B
3	25 - 06052	De Lumen, Janelle H.	NeonWebs

II. PROJECT INFORMATION

Project Title: ZooQuest

Project Background:

Many children cannot visit physical zoos due to geographic distance, high travel costs, or lack of local facilities. This can also affect the teachers because they cannot organize field trips and need accessible digital tools to teach biodiversity. But children have easy access to the internet and gadgets to communicate with one another. At the same time, they only spend hours on gadgets (mostly computers) consuming passive content (like short videos) and online games, which disconnects them from learning about nature and wildlife. This also affects their parents because they struggle to manage excessive, unhealthy screen time at home. This system will bridge the educational divide by granting every child, regardless of family income or location a free access to wildlife learning, while redirecting mindless screen time toward interactive mystery-solving that builds early environmental empathy and critical thinking skills.

Project Objectives

- **General Objective**
To develop a Java-Based System that is an interactive zoo mystery game that allows users to explore different biomes, learn about animals, and solve mysteries.
- **Specific Objectives**
The project aims to:
 - Develop an interactive virtual environment.
 - Enable user interaction with various animals in each biome; and
 - Integrate a clue-based mystery game mechanic.

Target Users

Target User	How Will They Use the System?
Children	Use the system to explore virtual zoos and their biomes, complete the task (find the animal), and learn their habitat, diet, and behavior.
Teachers/Parents	Use the System as an interactive learning activity to introduce



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	children to animals and wildlife.
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Proposed Features:

Feature	Description
User Interface	A user interface will feature a child-friendly, colorful theme, large button, simple icons, and clear navigation. It includes a menu page, biome map, achievements, and animal information panel.
Interactive Biomes and Animals	Virtual biomes such as Jungle, Safari, and Aquatic that children can freely explore, and each biome contains its own corresponding animals that children can view their habitat, diet, sounds, and behavior.
Mystery Game	A clue-based game in which children (user) visit biomes, observe animals, and collect clues to solve a wildlife mystery case.
Achievements	Badges or titles such as “First Clue Collected” or “Master Explorer” are awarded for exploring biomes, learning animal facts, and solving cases.

III. JAVA AND OOP CONCEPTS

OOP Concept	How It Will Be Used?
Classes and Objects	Blueprints for game elements and their running instances in memory.
Methods	For actions, behaviors, and event handlers of your objects.
Encapsulation	Protecting game state and data integrity by hiding internal fields.
Inheritance	Code reuse by building specific animal and zone types from common parent templates.
Polymorphism	Letting different objects respond to the same action in their own distinct way.
Abstraction	Hiding background mechanics and exposing only clear interfaces to the player and GUI.
Exception Handling	Preventing game crashes caused by missing files, bad user inputs, or invalid states.
Arrays/Data Handling	Storing, organizing, and traversing collections of biomes, clues, achievements, and animals.

IV. PROPOSED CLASSES

Class	Purpose	Main Attributes	Main Methods
Biome	A virtual biomes represent a habitat such as Jungle,	biomeName description isLocked	addAnimal() unlockBiome() getAnimals()



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	Safari, and Aquatic where children can explore	animalList biomeMystery	getBiomeMystery())
Animal	Represent a wildlife species that children (users) can interact with and discover clues.	name species habitat diet uniqueFeature imagePath associatedClue	displayBehaviour()) getDescription() getClue()
MysteryGame	Manages the mystery game/puzzle of a specific biome.	mysteryTitle targetAnimal clueList isSolved	getNextClue() verifyGuess() allCluesFound() solveMystery()
MysteryClue	Represent a single clue that tied to each animal that helps to solve the mystery case	clueId hintText isRevealed associatedAnimal Name	reveal() isRevealed() getHintText()
Achievement	A badge or title earned after exploring, learning, and solving the case.	badgeName badgeDescription iconPath dateEarned	viewBadge()
UserProfile	Stores player profile data, completed achievements, and overall unlocked progression.	username userAchievements userProgress	addUsername() editUsername() addAchievements() showProgress()
GameManager	Control overall game flow; loading biomes,	allBiomes currentPlayer currentBiome	initializeGame() processGuess() handleBiomeCompletion() saveProgress()
MainFrame	The primary desktop window that holds the whole application and manages switching between screens.	cardLayout containerPanel gameManager	showScreen() updatCurrentBiomeView() exitApplication()
MainMenuPanel	The landing screen where the user enters their name, starts the game, or checks trophies.	nameInputField startButton viewAchievement quitButton	mainMenuPanel() handleStartGame()) handleOpenAchievement()
BiomeViewPanel	Interactive overhead map screen allowing selection of unlocked ecological biomes.	biomeButtons progressLabel backButton biomeNameLabel backgroundDispl	refreshMapState() onBiomeSelected()) loadBiomeData()



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AnimalViewPanel	The main interactive playground where the user explores the animals.	animalGridPanel animalDisplay animalDescriptionText	createAnimalCard() onAnimalClicked()
MysteryGamePanel	The game section where the user plays the game, view (popup or drawer) where players read the clues, and submit who they think the mystery animal is.	mysteryTitleLabel animalDisplayList cluesDisplayList submitGuessButton closeButton currentMystery	displayAnimal() mysteryDialog() refreshClueList() handleSubmitGuess() displayResultFeedback()
AchievementViewPanel	The trophy room where kids can view all the badges they have earned across biomes.	headerLabel badgeContainerLayoutPanel backToMenuButton mainFrame	achievementViewPanel() refreshAchievement() createBadgeCard()
ZoneLockedException	Thrown when a player attempts to enter or skip ahead to a biome that has not yet unlocked	targetZoneName requiredZoneIndex	getTargetZoneName()
InsufficientCluesException	Thrown when a player attempts to make a final guess before discovering the minimum required number of clues in an animal	currentClues requiredClues	getMissingClueCount()
AnimalNotFoundExcption	Thrown when a player attempts to retrieve an animal by name, ID, or user query, but the entity does not exist in the active biome	searchedName biomeName	getSearchedName()



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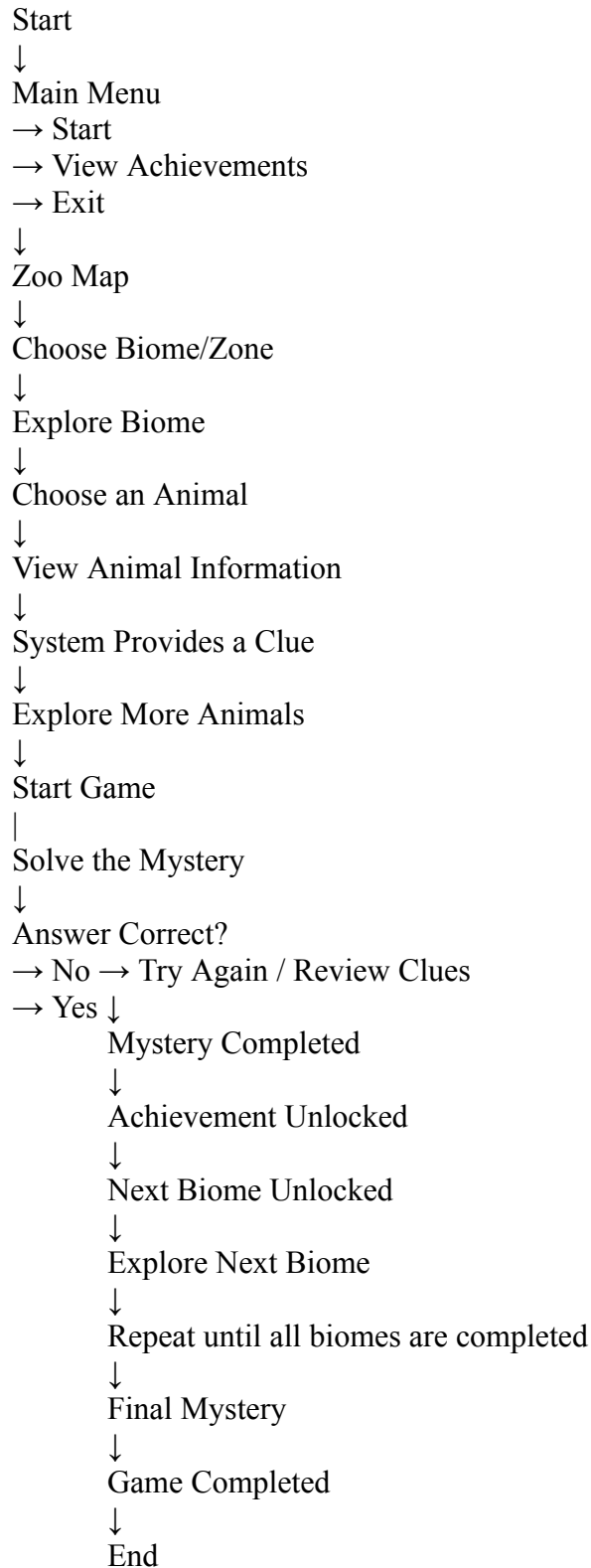
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V. PROGRAM FLOW





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VI. TEAM RESPONSIBILITIES

Members	Assigned Tasks
Arias, Jayro J.	- Develops system requirements, class designs, UML diagrams, and overall application structure. - Maintains the README, testing documentation, screenshots, and assists in checking the application's functionality.
Carenan, Ahron Simeon D.	- Coordinates the implementation of Java classes, methods, and OOP features - Develop the UI/UX design of the program
De Lumen, Janelle H.	- Develops the program flow and user interaction flow of the ZooQuest application. - Helps plan the GUI navigation and gameplay sequence, including biome exploration and mystery game progression.

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and Requirements	Project Proposal
Week 2	System and class design	Class/UML diagram
Week 3	Initial coding	Basic classes
Week 4-5	Feature development	Working features
Week 6	OOP implementation	Completed OOP features
Week 7	Testing and debugging	Tested application
Week 8	Documentation and presentation	Final project

VIII. EXPECTED OUTPUT

ZooQuest: A Java-based application where the user can explore different zones in a zoo and interact with various animals. Each zone has a mystery game, and the system will give the user a clue that can help solve the mystery. After getting enough clues and solving the mystery (find the animals), the user can unlock and move to the next zone until they finish the whole game.



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UI EXAMPLE



IX. PROJECT LIMITATIONS

- The system will only include a limited number of zoo biomes and animals.
- The game will only have pre-made clues and questions.
- The system will focus on basic 2D GUI interaction and will not include 3D graphics.
- The animal information and clues will be based on pre-set data stored in the system.

X. GITHUB REPOSITORY

Repository Name: ZooQuest

Link to Repository: [jayro26/ZooQuest](https://github.com/jayro26/ZooQuest)

